Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/18/2007
PI Name:	Bingham, Gail E Ph.D.		
Project Title:	Validating Vegetable Production Unit Plants, Protocols, Procedures and Requirements Using Currently Existing Flight Resources		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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City:	North Logan	**State:**	UT
Zip Code:	84341	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	03-OBPR-01
Start Date:	01/01/2004	**End Date:**	07/31/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: corrected end date to 07/31/2007 per PI (1/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Garland, Jay (Life Support Group, Kennedy Space Center) Bates, Scott C. (Department of Psychology, Utah State University) Wu, Yajun (Department of Plants, Soils and Biometeorology, Utah State University) Bugbee, Bruce (Department of Plants, Soils and Biometeorology, Utah State University)		
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA and the other space agencies of the Earth are considering including a Vegetable Production Unit (VPU) as a component of their space flight vehicles, including the Crew Exploration Vehicle (CEV) that is planned be used to carry humans from the Earth to Mars. As the US moves to define the CEV, critical questions about the role of a VPU and its optimum design and exploitation remain unanswered. Two critical questions revolve around the quantization of the "non-nutritional" benefits of plants in the cabin, and discovering what operational conditions maximize these benefits. These "non-nutritional values" have the potential to answer Critical Path Risks 24a and 24e, which carry priorities 1 and 2. If the presence of plants does provide significant non-nutritional benefit to long-term crews, and if part of the benefit is as a fresh food source, food safety issues become important. This effort will help addresses Critical Path Risks 10e and 10f, which carry Priority 1 and 3, respectively. In the process of monitoring microbial accumulations around the on-orbit plant growth facility (Lada), we will also explore Critical Path Risks 35a and 35d, both Priority 1 risks. Many of the Critical Path Risk 42 Lunar questions are also addressed by this work. The VPU plant, procedure, protocol, and food safety requirement (P3R) program is structured to secure this information on a schedule that will provide answers for the Spiral 2 and Spiral 3 CEV development efforts. Objective – Non-Nutritional Value: Quantify the mass value that should be assigned to the non-nutritional effects of plants in a spacecraft cabin. How much vegetable yield is required for a measurable boost to crew response under the isolation and stress of long-term space missions? How should this stress-relieving resource be presented (open or closed, single crew tended or multiple access) to maximize its value? Is this resource of value to all crew members, or is there a subset that prize the experience much more highly? Objective – Food Safety: Identify the threat levels and validate the procedures and protocols required to allow US astronauts to eat space-grown vegetables. Determine how to implement these procedures to maximize crew mental health benefits with minimum mission costs. What procedures and hardware need to be in place to assure crew health on a long flight? Objective – Plants: Identify, flight-qualify and optimize the support requirements for the vegetable and flower plant varieties that are most likely to be utilized to meet flight crew needs (wants) and that fit within the hardware resource limitations that exist on the ISS and will exist on the CEV. This work will validate the technology readiness level of our available cultural practices to provide reliable, low-cost, stimulating products for crew well being. What plants can tolerate cabin ethylene levels without adding significant cost? How well do seeds need to be protected to survive and flourish on the interplanetary trip? Will seed-producing plants be stable over the 500- to 600-day mission? The objectives are structured to provide the data necessary to determine whether VPU hardware should be considered for the CEV. It will also provide the data necessary to qualify plant products for authorized consumption on the ISS and the CEV by US crewmembers. The Lada hardware and our cooperative agreement with IBMP have been shown to provide ideal, continuing opportunity to investigate many of the plant, procedure, protocol, and food safety requirement (P3R) issues associated with meeting crew livability requirements.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Better understanding of the psychological value of flowers and plants. Better understanding of the threats to humans of greenhouse-grown plants.
Task Progress:	During the formulation phase of this project, we have demonstrated the feasibility of meeting all of the originally envisioned tasks and the ability to work closely between the two major ISS partners to secure the opportunities that were identified during our formulation efforts. For example: • We have begun the optimization of the VPU components to maximize utility and lifetime and minimize mass. This optimization includes ethylene sensitivity, plant nutrient uptake, and long term fertilizer studies to extend root module life. Identifying plant varieties that are ethylene tolerant but meet the nutritional and non-nutritional needs will be a significant advance, reducing cost and mass. • We have developed the plans and identified the resources to conduct a systematic study to examine the mix of vegetable and flower varieties required to provide maximum crew support value. The cultural practice parameter space will also be reviewed to determine what limitations the cabin environment places on production. • We are involved with IBMP in a study to determine whether a lightweight root zone substrate can be developed to reduce root module mass by more than 50%. Developing the cultivation techniques required to allow reducing the root module depth by half would reduce mass by another 25%. • We have received approval from the NASA Committee for the Protection of Human Subjects (CPHS) to administer a survey of space explorers regarding their thoughts and feelings about growing and tending plants during space-based missions. We have also submitted our protocols to the Utah State University (USU) Institutional Review Board and are preparing materials for the Russian Bioethics Commission and the Human Resource Multilateral Review Board (HRMR). • We have initiated a task to track and publish the genetic differences that accumulate during multiple generations of microgravity plant production. This effort will provide the world with the results of our 4 generation pea experiment completed recently on the ISS. We have also been invited to participate in the ESA/IBMP studies of DNA differences in multiple generations of seeds grown in space planned for the ISS in 2007. Cooperative work on these studies should allow us to participate on the higher orbit studies planned for later in this decade. • We have successfully demonstrated the use of existing ISS microbiological monitoring tools for plant assessment on the ground and in space. We have developed a cooperative effort with the IBMP microbiologists monitoring the ISS who will monitor in-flight loading on material grown in Lada. This effort will provide the operational NASA safety program with the data required to resolve the flight surgeon issues that are keeping US crew members from fully utilizing space vehicle produced plant products. • We held a team coordination meeting with our Russian team members and NASA Advisory Team in May 2006 at SDL to share results and to determine the technical and logistics details to be addressed in our continuing efforts. This team is ready to move forward into full fight status to conduct these studies.

	• We have established cooperative ties with the NASA-funded project Behavioral Issues Associated with Long Duration Space Expeditions: Review and Analysis of Astronaut Journals Experiment 01-E104 (Journals – Jack Stuster), to amplify benefits from both efforts to understand psychological well being and the green plants in isolation. Proceeding with this program would allow us to cooperate more closely with this study, adding benefit to both programs. • We have established a cooperative psychological program with the IBMP scientists studying the stress and behavior modifications required to maintain mental health and performance in the confined environments of space flight. We can utilize access to the Russian crews for survey collection, both on the ground and (potentially) in space to evaluate plants and their presentation for non-nutrition benefits. Through our Russian colleagues, will be able to look at their air-to-ground logs that refer to the plant studies, and we will be added as co-authors to jointly publish the results of this program. • We have been invited to support (as Russian team members) the ESA/RSA Mars/500-Day Chamber Study to be conducted at IBMP during 2008. This experience will provide a method of evaluating VPU design optimization data that cannot be collected in any other way. It will allow us to minimize the small number issue with previous studies by allowing us to make multiple presentations of the hardware and plants to the same crew, monitoring results with high level asset approach and survey tools not available on the International Space Station (ISS) or in shorter studies.
Bibliography Type:	Description: (Last Updated: 03/08/2018)
Papers from Meeting Proceedings	Bates SC, Marquit JD. "The benefits of plants: a review of the literature and application to habitation systems for humans living in isolated or extreme environment." Conference on Habitation Research and Technology Development, Orlando, FL, February 2006. Conference on Habitation Research and Technology Development, February 2006. , Feb-2006
Presentation	Roberts, M. S.; Hummerick, M.; Bingham, G. E.; Garland, J. L. "Development of a quantitative microbial risk assessment for Lada, an ISS plant growth system" 3rd International Workshop on Space Microbiology May-2006